



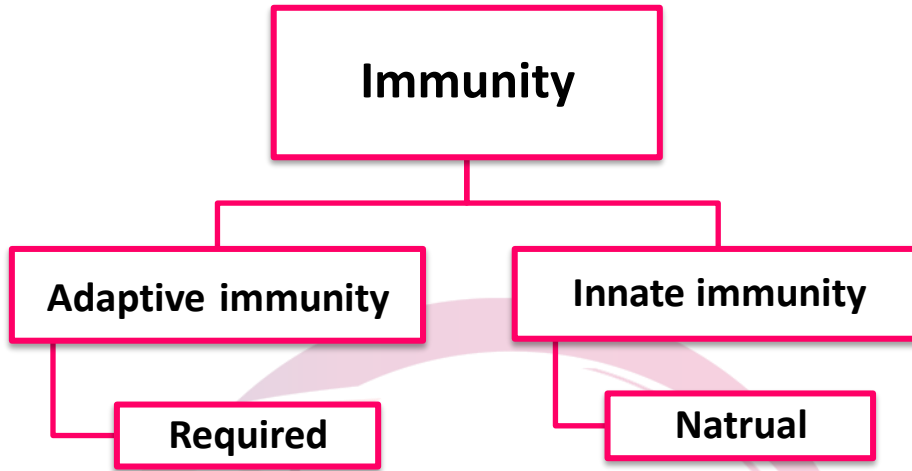
2020

**Lab notes and
procedures**

GENETIC ENGINEERING &
BIOTECHNOLOGY
DEPARTMENT

Immunology & Sereology LAB

LAB#1 (IMMUNE SYSTEM)



Adaptive immunity المناعة المكتسبة	Innate immunity المناعة الطبيعية (فطرية)
Specific	Non specific
Memory	Not memory
Need for expose to foreign substances	No need for expose to foreign substances
Eg:(lymphocytes) B-cells(humoral) T-cells(cellular)	Eg: tears, skin, gastric juice, Phagocytes(macrophages ,neutrophils ,natural killer cells)

المناعة الفطرية (الطبيعية): يتكون موجود بالكائن الحي عند الولادة أو حتى في المراحل الجنينية الأولى



المناعة المكتسبة: المناعة يكتسبها كل انسان بمفرده حسب المرض يلي بهاجم الجسم ولهاد السبب هادي المناعة ببيكون الها ذاكرة memory يعني بقدر يتعرف على أكبر عدد من مسببات الامراض



★ **Antigen(Ag):** Foreign substance(molecule)that could enhance immune system or couldn't enhance immune system.

★ **Immunogen:** foreign substance that must enhance immune system.



ركزوا على must

➡ **Note:** all immunogen are antigen, but not all antigen are immunogen

★ **characteristics of antigen:**

1.must be protein (high molecular weight)

★ **characteristics of antigen to be injected:**

2.free of toxic

3.steriled(no microorganism)

4.pure (one epitope)

5.suitable PH

6.dose(low dose, suitable dose, high dose)

Epitope :small site on an antigen that is recognized by the immune system

#LAB2(IMMUNIZATION)

Dose: amount of antigen per kg of animal

-**high dose** :as self Ag and tolerance (no immune response).

-**low dose**: no immune response.

-**suitable dose**: strong immune response.

★ Routes of admission for Ag into body:

1.Nose+ mouth

2.Intradermis(I.D):inside skin

3.Intraperitoneal(I.P):inside peritoneal cavity

4. Intravenous (I.V):inside vein

5.Intramuscular(I.M):inside muscles

6.Subcutaneous(C.S):under skin



Intraperitoneal

حقن داخل التجويف البطن وهادي أضمن طريقة لما
نعمل injection ل mice (شوفوا الصورة)

★ Animals:

1.Mice 2.Rat 3.Rabbit 4.chicken 5.Hamster

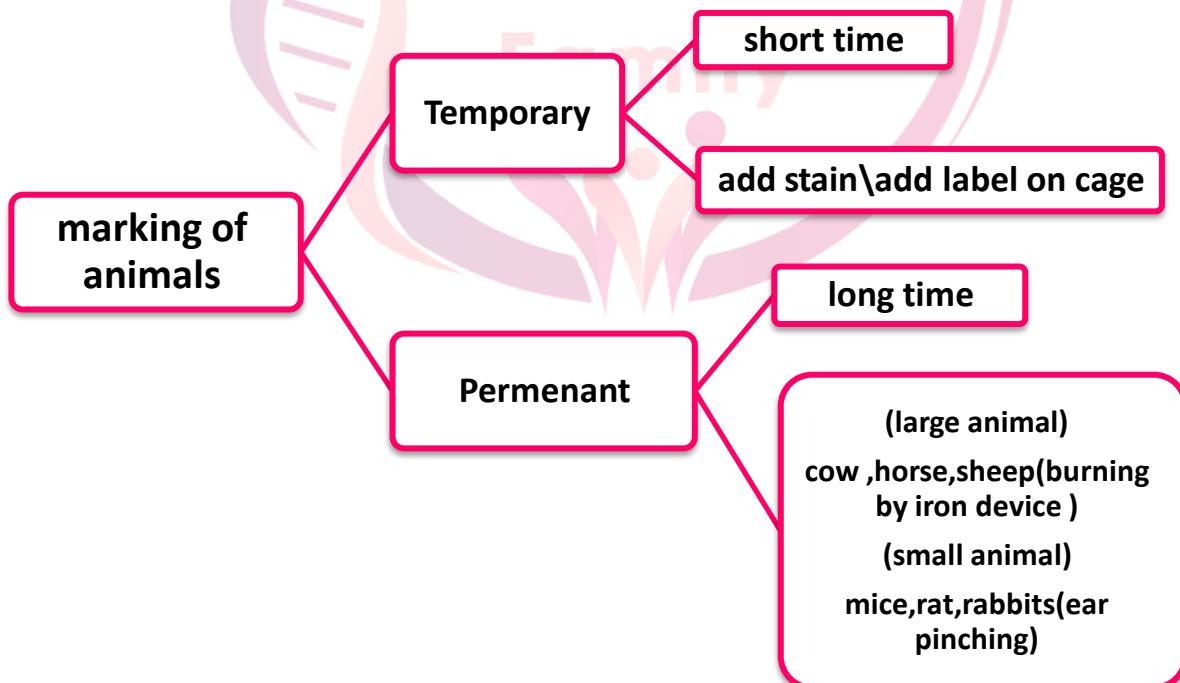
★ Mice:

Bulb c, white mice :1.98% genetically identical

2.interbreeding between sisters and brothers more than 20 times.



★ Marking of animals:





★ Types of Antigens:

Soluble(need adjuvant)

*Bovine serum albumin(BSA)

*Transferrin

Particulate(no need for adjuvant)

*RBC, Bacteria

Transferrin

عبارة عن بروتين سكري يربط خلايا الدم بالبلازما / ومسؤول عن نقل الدم الى الخلايا مثل الكبد الطحال.

Bovine serum albumin:

عبارة عن بروتين يتم استخراجه من مصل اللبن من البقر ويحتوي على نسبة عالية من الاحماض الامينية

★ **Adjuvant**: substance add to only soluble Ag to all route of admission except Intravenous :to do:

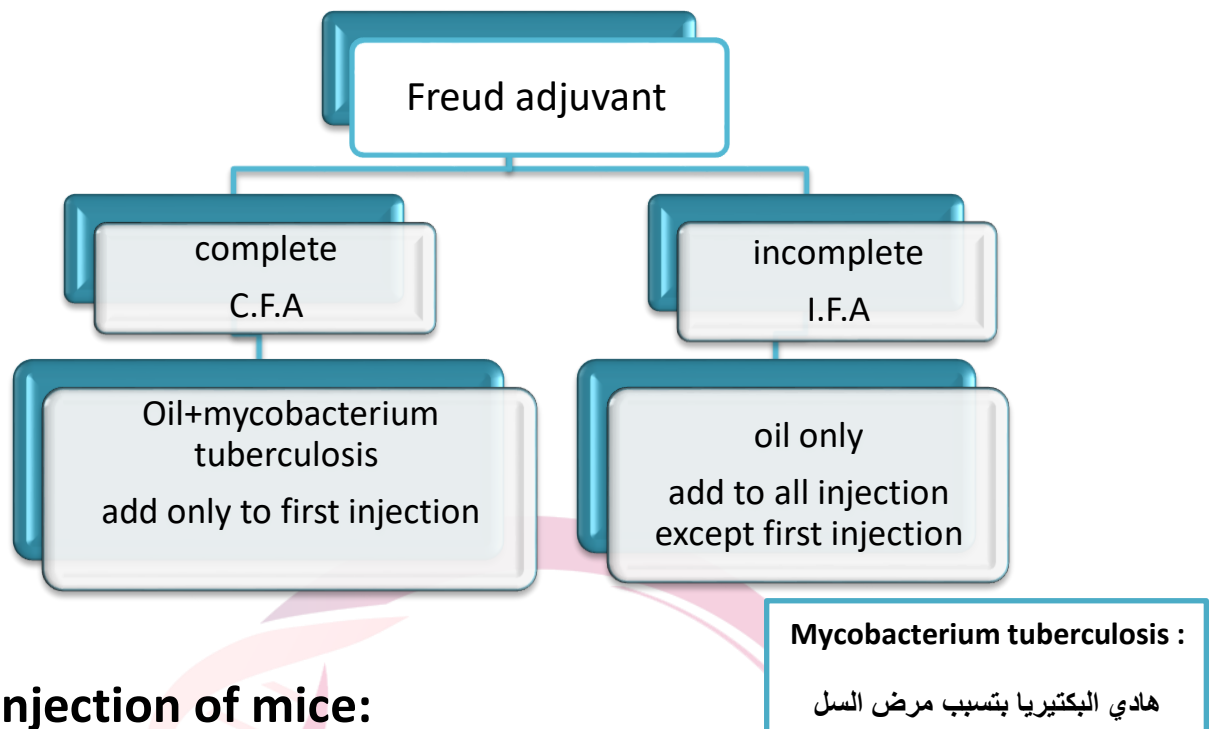
1. slow release Ag to tissue

2. slow absorb Ag by tissue

3. to prevent immune shock+ enhance immune system induction

Adjuvant:

مادة مساعدة بنضيفها مع اللقاح / ليش؟ حتى يزيد من الاستجابة المناعية



★ Injection of mice:

(BSA)*protein derived from cow*

#Ag bovine serum albumin with complete freud adjuvant

• Dose:

Dose of B.S.A: 50Mg-100Mg/mouse

100 µl BSA : 100 µl C.F.A → 200µl /1 mouse

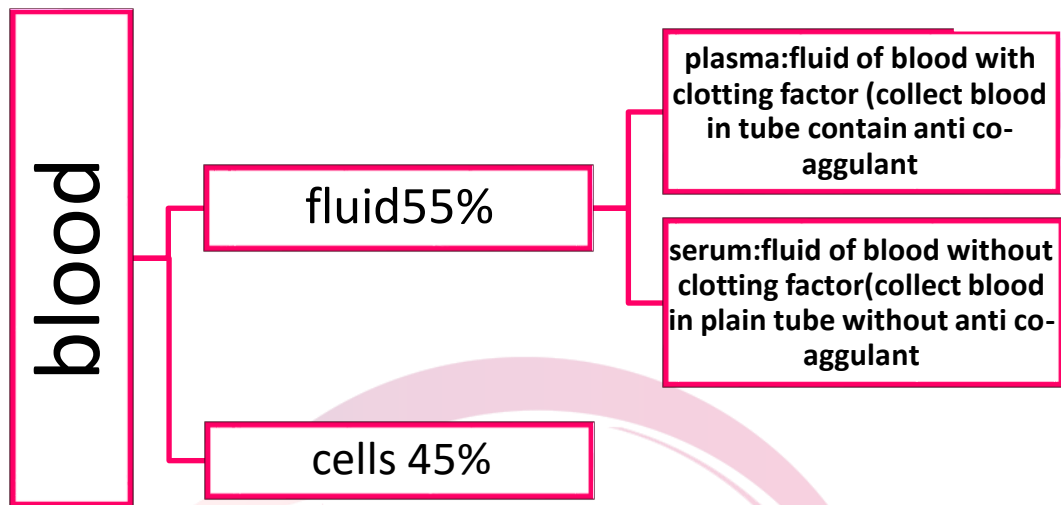
C.F.A:B.S.A
1 : 1

→ Route of admission (Intraperitoneal injection)

Note: we do this experiment to have Abs against B.S.A



LAB3#(BLOOD)



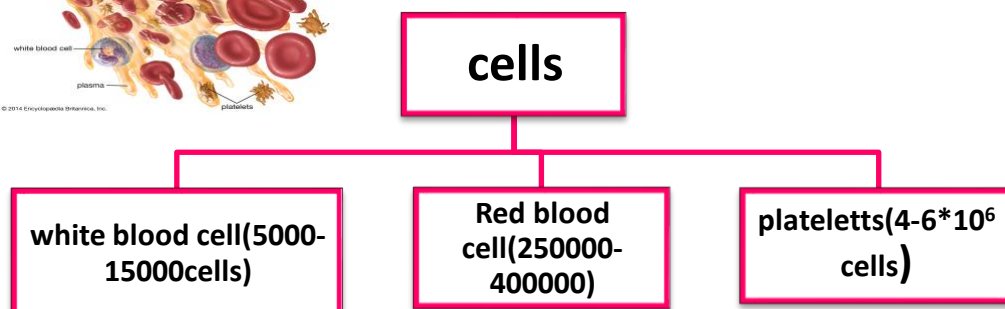
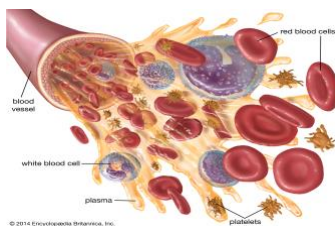
★ **Serum tube:** plain tube with red cap



→ **Types of anti-co-aggulant:**

1. EDTA (with violet cap) 2. Ca oxalate 3. Na citrate
(attack (bind to Ca^{+2}))

4. Heparin (with green cap): in tissue culture.



White Blood Cell have two
types:1.Granulocyte/2.Agranulocyte

Granulocyte

neutrophile(65-75%)

Eosinophile*red granules*(0-1%)

Basophils*blue granules*(0-1%)

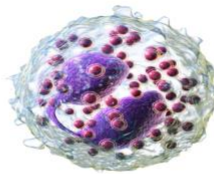
Agranulocyte

monocyte(the largest)(3-8%)

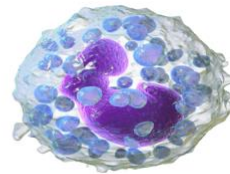
lymphocyte(25-35%)(B-cells/T-cells)



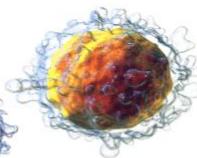
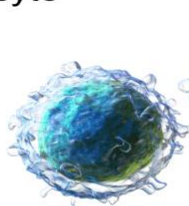
Monocyte



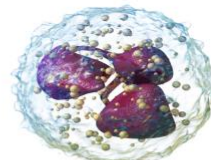
Eosinophil



Basophil



Lymphocytes



Neutrophil

White Blood Cells

★ Count of WBC:

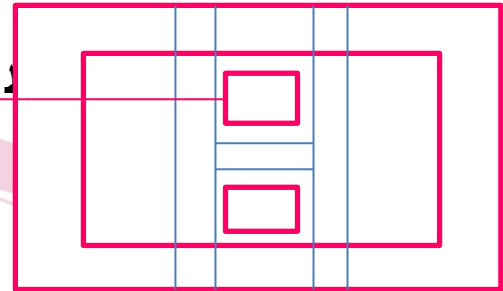
Neubauer chamber(Hemocytometer)

الاسم كثير مهم ورح يجيكم
سؤال بالامتحان

W:count large cells(WBC ,macrophage ,fungi)

R:count small cell(RBC, yeast, bacteria, sperm cells)

W		W
	R	
W		W



W:contain16 small squares

Count at any corner start= zigzag lines

Calculation:

#of cells in $W \times V \times CF \times DF = \dots \text{cell/mm}^3$

DF: final V/sample V $\rightarrow 950\mu\text{m (acetic acid)} \times 50\mu\text{m}$
sample/50 μl

V.CF: volume* conc.factor

Function of acetic acid: lysing RBC

★ Procedure:

1.collect blood in EDTA tube(violet cap)

2.950 ml acetic acid add 50 ml whole blood in eppendorf tube,
mixing until brown color of sol appear(lysing)



3.prepare hemocytometer clean it by tap water ,then dry by lab coat

4. clean microscope by lens paper

5.put cover slip on hemocytometer to see lines under low objective lens 4x

6.count white blood cells (shine &dark)at medium (10X)

7.calculation

★ Blood smear (film):

1. To see shape of RBC, volume, color 2. See some parasite

3.see the shape of platelets

4.count for different shape of WBC(100 cells)

★ Procedure:

1.take whole blood(with anticoagulant),put few drops on slide.

2.spread blood on slide like (tongue shape),then leave for air dry

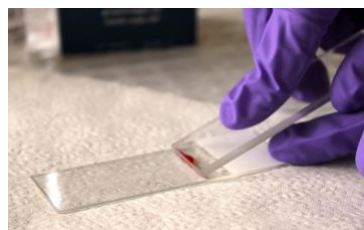
3.add stain for one min (Right stain)

4.wash by tap water ,leave dry, then see under microscope(40X)

خطوة 2 جدا مهمة ولازم نشتغلها صح لأقدر أشوف العينة بعد ما أصبغها وأكون مميزة شو يشوف ,ف لما تعمل spread خلي ايدك خفيفة ليطلع معك tongue shape

Right stain:

- 1.acidic stain :red Eosin stain
- 2.basic stain: blue hematoxylin



LAB#4(Serum Preparation)

First step:

Fibrinogen(inactive form)/soluble $\xrightarrow{\text{Ca}^+ / \text{enzyme}}$ **fibrin**(active form thread) /insoluble *15-30 min*

★ Ways to take blood from animals:

1.during survival(still a live):how:?

- a. Retrioorbital puncture(vein of orbit of eye)(blood capillary).
- b. Cardiac puncture(from heart(high amount)).
- c. Vein puncture (tail: mouse)/(ear: rabbit).

2.Decapitation(Dead):(remove from head):

-Need high amount of blood.



★ Steps for serum preparation:



- 1.coagulation clotting(blood in plane tube(red cap)
- 2.centrifugation at 3000 Rpm for 5 min ,then transfer serum by glass pipette eppendorf tube
- 3.clearing:clear serum by remove fat: (fasting animal for 12 hr, ultra centrifuge 25000 Rpm(15 min at 4 c).
- 4.preservation:adding of preservative: to stop cell division of microorganism how?

1-methiolate/ 2-phenol/ 3-sodium azide

5.storing:

- 1.serum keep for long time at -20°C
- 2.analysis IgE antibodies keep at -7°C
- 3.powder serum 4°C (liquid-solid)

#ammonium sulfate(NH_2SO_4):more purification of protein (Ab)

4/5 مهمات كثير ويبجي عليهم أسئلة
كل سنة

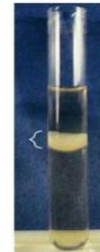
LAB#5(precipitation test)

1.Double ImmunoDiffusion (D.I.D)

2.Single Radius Immune Diffusion

3.Immuno electrophoresis

● soluble Ag+ soluble Ab $\longrightarrow$ ppt test



● large particles Ag +soluble Ab $\longrightarrow$ Agglutination



★ Factors affect Ag-Ab complex:

- 1.concentration of Ag-Ab
- 2.physical factors(PH, temperature ,pressure)
- 3.affinity of Ab (strength between Ab and Ag)

★ Reaction between Ag and Ab zones:

1.pre-zone:

$[Ag] \ll [Ab]$ / -There is no white ppt result

2.post zone :

$[Ag] \gg [Ab]$ / -There is no white ppt result

3.Eq-zone:

$[Ab] \approx [Ag]$ -There is white ppt result (visible)

📌 ركزوا هون (مهم جدا)

★ Double immunodiffusion :(ouchterlony test)

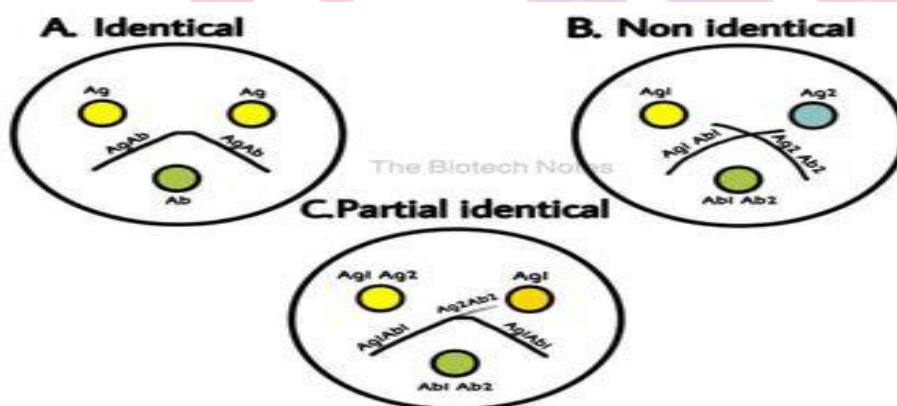
- Double: Ag and Ab move to each other
- Immuno: Ag-Ab Rxn
- Diffusion: principle of techniques (moving from high concentration to low concentration)

★ Application:

- 1.test for homogeneity between Ag and Ab
- 2.test for purity of Ag (serum)
3. test for semiquantitative(comparison)

رکزوا ہون کو پس

★ Types of reaction : ppt in DID:



★ 1. Identical Rxn :

If white ppt lines are fused together ,so Ag1 is identical to Ag2

★ 2. Partially identical Rxn :

If white ppt lines are fused together with form ,so Ag1 is partially identical to Ag2

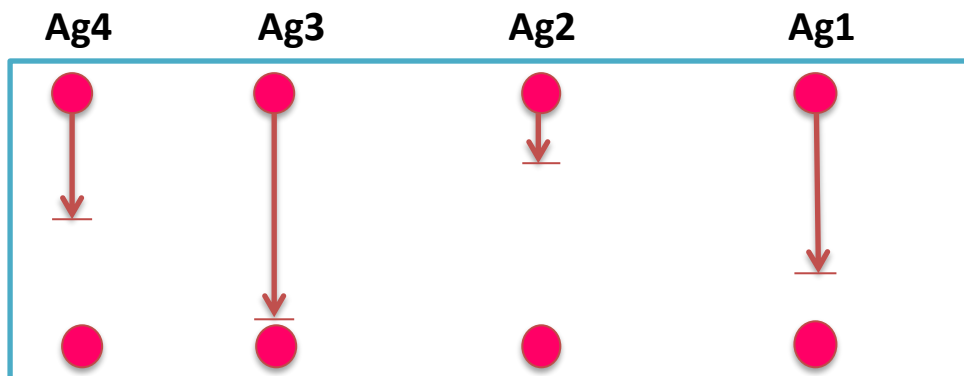
★ 3. Non identical Rxn:

If white ppt lines are crossed, Ag1 is not identical to Ag2

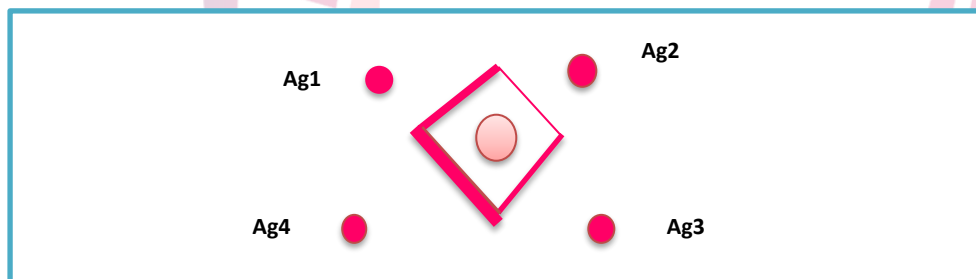
★ Semi quantitative:

*distance travel by Ag or Ab, Ab , concentration of Ab or Ag 

*thickness of ppt line



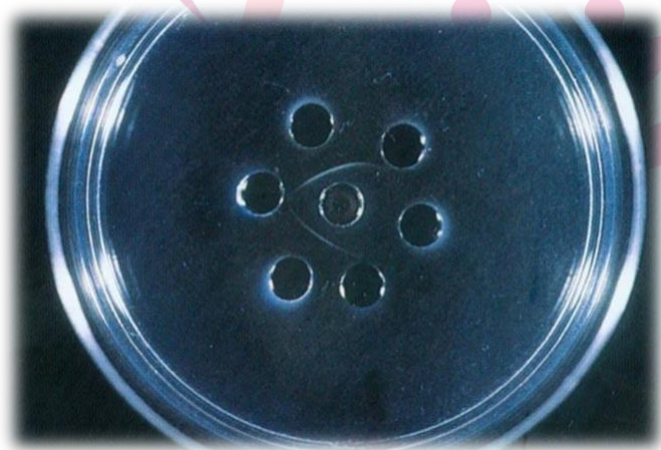
$(Ag3) > (Ag1) > (Ag4) > (Ag2)$



$Ag4 > Ag1 > Ag3 > Ag2$

★ Procedure:

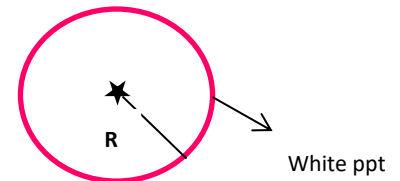
1. precoating slide(with 1% agarose)
- 2.add 5-6 ml melting agarose at (53 c), 2% and leave slide for 15-30min(to be solid)
3. add adaptor on surface of agarose(to make wells)
4. remove agar from wells by tip
- 5.add 10 ml anti mouse IgG (Ab)
- 6.add 10ml mice serum around
- 7.incubate slide at humid chamber (at room temp)for 24-48 hrs.
8. see your result





LAB#6 SRID /Mancini method

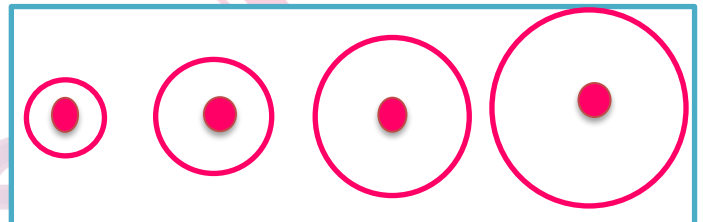
- S:Single(one of them: Ag or Ab)
- R:Radius(ppt hallow circular radius)
- I:Immuno: (Ag-Ab rxn)
- D:Diffusion (higher ions)



★ Application:

1. measure concentration of unknown Ab and Ag)
- 2.quantitative

diameter ↑ [Ab] ↑ [Ag]



★ procedure:

- 1.pre coating slide with 1%agarose
- 2.add 100 ml anti mouse IgG with melting agarose(5-6 ml) or add 100 ml anti human with melting agarose(5-6 ml)
- 3.leave slide for 15-30 min to be solid
- 4.make serial dilution for mouse serum or human serum
5. make 5 wells on the same line and remove agar by tip
- 6.add 10 ml from each dilution in each well in ascending or descending
- 6.add 10 ml from each dilution in each well in ascending or descending
7. put slide in humid chamber/see your result after 24-48 hrs.

Lab#7(Immuno electrophoresis)

- Immuno: Ag-Ab reaction
- Electrophoresis: separation of substance by electricity due to charge

★ Immune electrophoresis has two method:

1.electrophoresis 2.D.I.D(Double immunodiffusion)

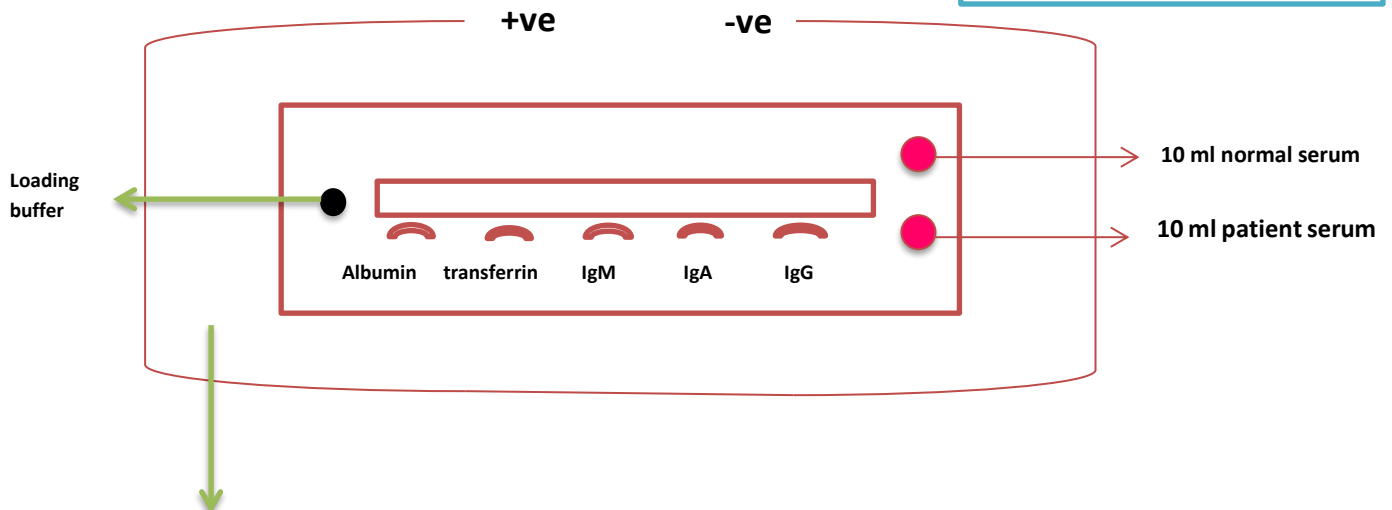
★ Applications:

1. separation of Ag or serum mixture
- 2.seperation due to charge
- 3.semiquantitative for immunoglobulin(IgG,IgM,IgD,IgE)

★ I.E has advantages over D.I.D:

1. less chance for over lapping
2. separated of mixture due to charge
- 3.used for clinical significant related to blood or Ig.

ركزوا مع المس مزبوط بهاد الشكل
لتقدروا تفهم



(Tris-HCl: pH=8.6), cover 90% of proteins running for e with -ve charge .

1. So sample add near -ve and move to +ve)

2. 1-2 cm for zero charge or slightly -ve charge (left one-two finger space)

3. add 10 ml of different sample on each wells

4. 100 ml anti human IgG (Homogenous Ab) / anti human serum (heterogeneous)

★ *100 ml of anti-mouse IgG (Homogenous) / anti mouse serum (heterogeneous)

5. (T=35-40 min / V=120 v)

يعني منشغل الجهاز ل 40-35 دقيقة / 120 فولت

6. albumin -4 / transferrin -3 / IgA -2 / IgM -1 / IgG 0

★ *albumin -ve (highest negative charge)

وبما انها اعلى شي يعني يتمشي بسرعة ل القطب الموجب

★ Note:

● if one white ppt are appears : homogenous Ab (one type)

● if two or more than one white ppt are appears : heterogeneous (2 or more type)

الصفحة لبعدها ترتيب
وملخص خطوات لهادي ال
procedure

★ procedure:

1. precoating slide with 1% agarose
2. 5-6 ml melting 2% agarose
3. leave slide 15-30 min to be solid
4. make two wells and between them (left one finger space)
5. add 10 ml in each well from different sample (normal / patient)
6. add 10 ml of loading buffer on one sample
7. put slides in electrophoresis at -ve electrons
8. add tris-HCl, pH=8.6 to counter
9. run electrophoresis for 35-40 min at 120 v
10. remove over from through
11. incubate slides at humid chamber (at room temp for 24-48 hrs)
12. see your result

loading buffer: glycerol+
bromophenol blue

function: 1. glycerol: decrease
the density of sample
/ 2. bromophenol: tracking of
electrophoresis

★ **Test bank:**

1. which one of the following is part of innate immune system:

- a. Immunoglobulin **b. Macrophages** c. B- lymphocyte d. T- lymphocyte
-

2. the relation between antigen & immunogenic protein is:

- a. Each immunogenic protein are antigen**
b. Each antigen protein are immunogenic
c. antigenic protein are the same as immunogenic protein
-

3. cardiac puncture is taken blood from :

- a. Vein b. Ear **c. Heart** D. head
-

4. one of the following is not true regarding antigen used in immunization:

- a. must be highly pure b. suitable dose
c. sterile **d. dissolving in DMSO**
-

5. The best way permanent mice tagging by :

- a. Ear pinching** b. branding
c. staining d. all of the above
-

6. adjuvant are added to antigen in case of all administration routes except:

IV: Intravenous

- a.IM b. IP c. IV d.SC

7. The best temperature for long term preservation of serum is:

- a. 4 °c b. room temp c. 37 °c d. -20 °c

8. factors that effect Ag-Ab complexes are: select one:

- a.PH b. concentration of Ag &Ab
c. temperature d. all of them

9.the visible precipitation in double immunodiffusion test occurs at the:

- a. post zone b. pre zone c. Equivalent zone

10.If mouse is injected IP with 10 mg BSA protein in complete : Freund adjuvant, the expected immune response will be

- a. very high b. moderate
c. tolerance d. fatal anaphylactic shock

11.single radial immunodiffusion test:

- a. is quantitative
- b. Qualitative
- c. is semi quantitative

12.precipitation line in double immunodiffusion test mean:

- a. reaction between soluble Ag & soluble Ab
- b. reaction between soluble Ag & insoluble Ab
- c.reaction between large Ag & insoluble Ab
- d. reaction between large Ag & soluble Ab

13.in double immunodiffusion test ,thick precipitation line between Ag &Ab means:

- a. high positive charge of the Ag
- b. low positive charge of the Ag
- c. high concentration of the Ag
- d. low concentration of the Ag

14.if two precipitation lines are crossed in double immunodiffusion test means that :

- a. two Ags are identical
- b. two Ags same epitopes
- c. an artifact (error)
- d. two Ags are non-identical

11.a

12.a

13.c

14.d

15.The following I correct about single radial immunodiffusion test:

- a. Manicini test
- b. an Ab concentration can be quantified
- c. is a type of immunogglutination test
- d. more sensitive than immune electrophoresis

16.when soluble antigen diffuse in a gel that contain antibodies, the optimum precipitation occur in.....

Ans : zone of equivalence

Optimum precipitation:

الترسيب الأمثل / الاوضح

17.if you want to compare two protein mixture which test you will done:

- a. immunoelectrophoresis b. manicini
- c. ouchterlony d. a and c

18.Double immunodiffusion test is a semi quantitative test:

- a. true b. false

19.counting of WBC can done by specific slide called.....,and acetic acid was added to blood before counting to

Ans: hemocytometer / lysis RBCs

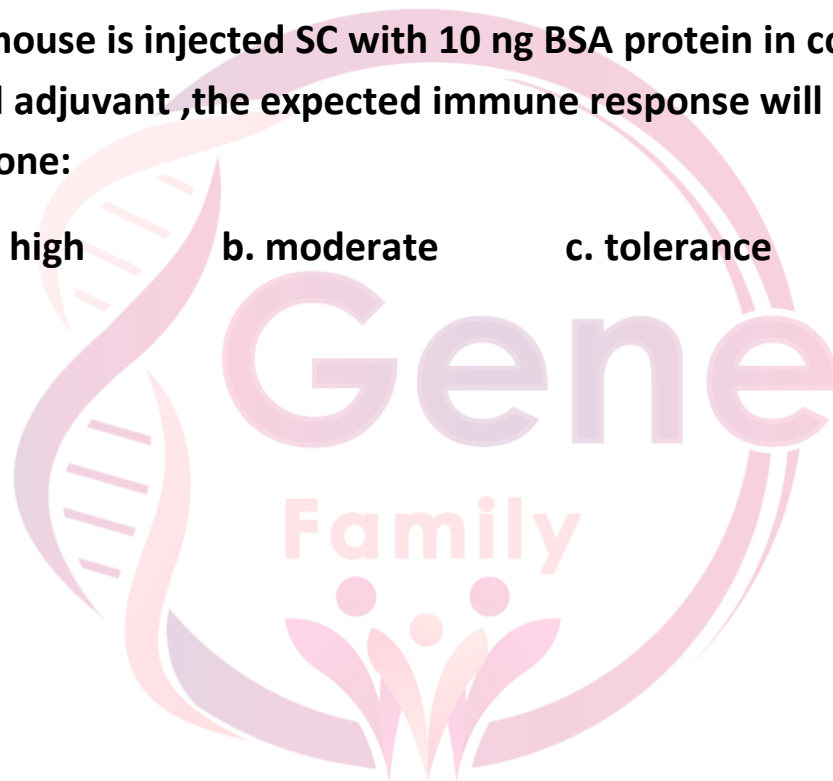
15.a 17.d 18.a

20. In single radial immunodiffusion test: select one:

- a. Ag &Ab are mixed with agarose**
- b. both Ag &Ab diffused toward each other**
- c. the Ag is mixed with agarose and the Ab is the unknown**
- d. the Ab is mixed with agarose and the Ag is the unknown**

21. if mouse is injected SC with 10 ng BSA protein in complete Freund adjuvant ,the expected immune response will be, select one:

- a. very high**
- b. moderate**
- c. tolerance**



20.d

21.c

DONE BY:JULNAR AYMAN